

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26171
- **Problem Statement Title** – On-device Visual Perception for Light-weight Browser Agents
- **Theme** – Smart Automation
- **PS Category** – Software
- **Team ID** – 124701
- **Team Name (Registered on portal)** – NEXORA_

AEGISBROWSE

Privacy-Preserving Visual Browser Agent

Video walkthrough: youtu.be/heZiO6_-ZtE



Proposed solution and how it addresses the problem

A local privacy layer between the screen and the server

1 Observe privately

Raw screenshot + DOM



Transport Department eSeva

Applicant Name: Rohit Khan

Licence Number: MH29 20251234

Vehicle Number: TN20 ZS 4814

Correspondence will be sent to the address below.
233, Gandhi Marg, Kochi - 440369

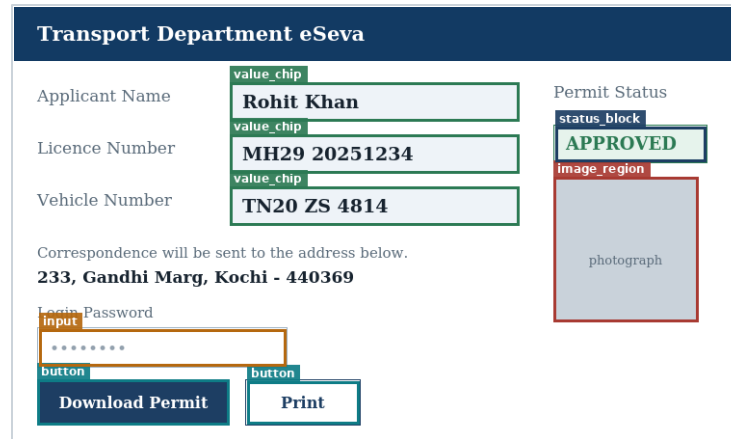
Login Password: [REDACTED]

Permit Status: APPROVED

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2 Locate sensitive regions

Structural + visual signals



Transport Department eSeva

Applicant Name: Rohit Khan

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3 Prepare safe context

Redacted pixels + metadata

Synthetic portal illustration



Transport Department eSeva

Applicant Name: [PERSON_1]

Licence Number: [LICENCE_1]

Vehicle Number: [VEHICLE_1]

Correspondence will be sent to the address below.
[ADDRESS_1]

Login Password: [REDACTED]

Permit Status: APPROVED

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CAPTURE AND REDACTION STAY ON THE DEVICE

Then select only task-relevant context for the server

Innovation and uniqueness

Fuse privacy signals

One mask for pixels and text

Escalate selectively

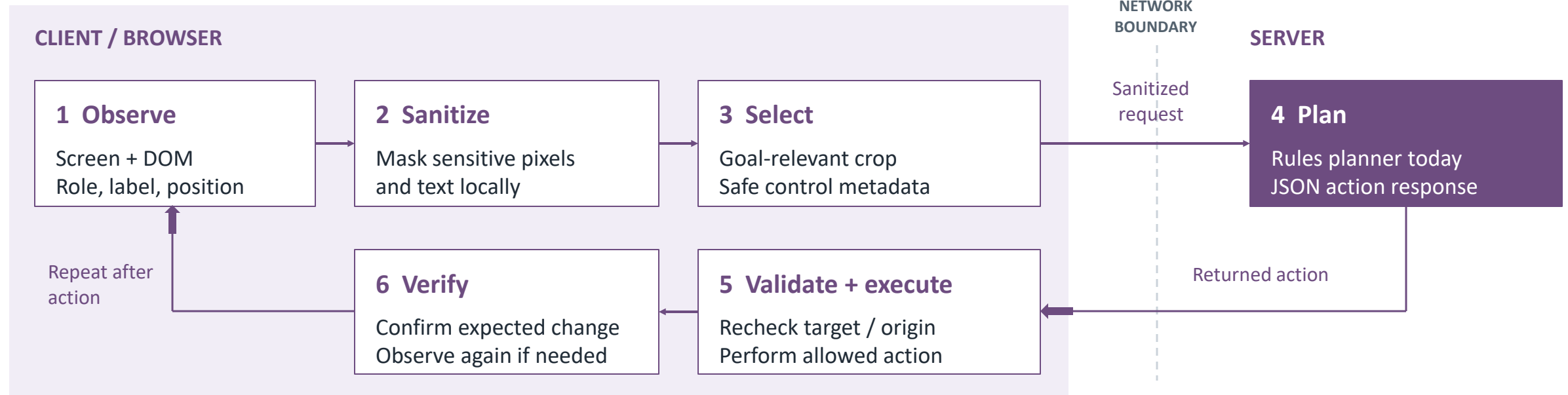
OCR where DOM text is absent

Disclose by task

Send only the useful context

Technologies and implementation approach

The browser protects data and controls every action



Demo stage

Chrome MV3 extension + FastAPI rules server
DOM privacy, goal crop, guarded action and verification

ONNX Runtime Web: integration scaffold

Optional loader and fallback are in place.
Real runtime assets + trained detector are pending.

Target visual path: ONNX Runtime Web with WebGPU / WASM. Open-weight VLM and Firefox validation are next milestones.

Analysis of feasibility

48 fresh synthetic pages / 4 portal families / 408 sensitive annotations

Criterion and measure	Always-on	Selective
Visual context (25%) UI F1 @ IoU ≥ 0.5	0.924	0.924
PII detection (20%) precision / recall	0.937 / 0.955	1.000 / 0.958
Redaction (20%) pixel precision / recall	0.643 / 0.977	1.000 / 0.973
Resources (20%) process RAM / CPU per page	57.3 MB / 52.1 ms	56.4 MB / 25.7 ms
Latency (15%) sanitize-to-plan median / p95	339 / 467 ms	47 / 48 ms

CPU reference harness, Intel Core Ultra 9 275HX: classical CV + rules planner. UI F1 is not mAP.

Python-process RAM excludes Tesseract child memory. Timing excludes browser execution and verification.

Challenges and improvement strategy

DOM-blind recall: 0.878. Improve OCR with field labels. Container rejection already reduces whole-card masking.

Still not measured

1 Trained ONNX detector

Accuracy on held-out screens

2 WebGPU timing in the browser

Cold / warm inference and memory

3 Whole-task time inside the browser

Observe through verified completion

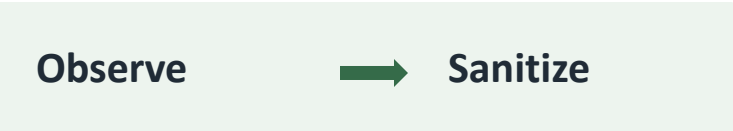
Potential impact on the target audience

Private assistance for authenticated portal tasks

Government services, scholarships, banking KYC, health records and employee self-service.

Example: download a certificate or report

01 Protect on the device



Screen + DOM; local privacy filter

02 Plan from safe context



Filtered request; choose Download

03 Act and confirm



Guarded click; confirm download

Measured on 48 synthetic pages: sanitize, send and select a control

Demo flow; completion timing pending

Benefits of the solution

PRIVACY

0

PII strings in sent metadata

SPEED

7.3×

Faster sanitize-to-plan median

COMPUTE

51%

Lower Python CPU time / page

PLANNING

48/48

Correct Download target chosen

Details / links of reference and research work

Mind2Web	Generalist web-agent benchmark arxiv.org/abs/2306.06070
SeeAct	Grounding web actions arxiv.org/abs/2401.01614
OmniParser	Visual screen parsing arxiv.org/abs/2408.00203
PrivWeb	Local privacy filtering and controls arxiv.org/abs/2509.11939
WebPII	Visual PII benchmark and detector arxiv.org/abs/2603.17357

Prior work and our contribution

- 1 Established foundations**
Local privacy filtering + visual interface parsing
- 2 Our evaluated change**
Reject content containers as photos; run OCR only where DOM text is absent.
- 3 Next evidence**
Measure the trained browser model and isolate the benefit of task-aware context.

Our research work: synthetic Indian-portal benchmark

12 dev pages

Scholarship + HR

➡ **48 fresh test pages**

4 held-out families / 408 items

➡ **48 stress pages**

DOM text deliberately removed

Benchmark evidence: nexora_benchmark/SELECTIVE_RESULTS.md + stored JSON results. Ground truth is used only by the scorer.